

0 DISCUSSIONS

Choose only ONE best answer.

D THHN

This answer is correct.

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0 DISCUSSIONS

QUESTION 2 OF 102

A drawing that shows the layout of a building's electrical systems, including the vertical paths of electrical wiring.

Choose only ONE best answer.

A

Electrical layout

B

Riser diagram

✓

C

Electrical plan

✗

D

Electrical master plan

This answer is incorrect. The correct answer is 'B'.

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A tobacco farm has its farthest load located 310 linear meters from a single phase power source. System voltage is 230 volts. Excluding bends, what wire length should be used in the voltage drop computation?

Choose only ONE best answer.

A310 m

B930 m

C620 m

D155 m

This answer is incorrect. The correct answer is 'C' .

Wire length = Distance x 2 wires/line
Wire length = 310 m x 2 wires/line
Wire length = 620 m

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0 DISCUSSIONS

Determine the reliable power output of a 12-volt DC motor if it is rated at 9.77 Amperes. Assume a system efficiency of 82%.

Choose only ONE best answer.

A

120 Watts

B

96 Watts

✓

C

690 Watts

D

600 Watts

This answer is correct.

Form the theoretical formula $P = VI$, multiply VI by efficiency to get output

$$P_{\text{output}} = \text{Voltage} \times \text{Amperage} \times \text{System efficiency}$$
$$= 12 \text{ V} \times 9.77 \text{ A} \times 0.82$$
$$= 96.1 \text{ Watts}$$

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Determine the power input of a 24-volt DC motor system having a total amperage of 20 Amperes. Use 82% system efficiency.

Choose only ONE best answer.

A120 Watts

B96 Watts

C690 Watts

D585 Watts

This answer is correct.

Form the theoretical formula $P = VI$, divide VI by efficiency to get input

Power input = Voltage x Amperage / System efficiency

= 24 V x 20 A / 0.82

= 585.37 Watts

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QUESTION 6 OF 102

0 DISCUSSIONS

Quantify the surge current of a 24-volt DC motor if it is supplying current to a 528-Watt, 24-Vdc sprinkler. Surge current is assumed at 25%.

Choose only ONE best answer.

A

27.5 A

B

27.2 A

C

2.27 A

D

22 A

This answer is correct.

$$P = VI$$
$$I = P/V$$
$$= 528/24$$
$$= 22 \text{ Amp}$$

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B27.2 A

C2.27 A

D22 A

This answer is correct.

$$P = VI$$
$$I = P/V$$
$$= 528/24$$
$$= 22 \text{ Amp}$$

Since surge current is approximately 25%

$$\text{Surge Current} = (1 + \text{Surge current in ratio}) \times \text{Load amperage}$$
$$= (1 + 0.25) \times (22 \text{ Amp})$$
$$= 27.5 \text{ Amp}$$

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QUESTION 7 OF 102

You need to do the planning of an automated control for the switching on and off of the electricity supply of an AC powered rain gun type sprinkler system. What is the appropriate central controller of your system?

Choose only ONE best answer.

AComputer

BIrrigation controller

CCentral pivot controller

DMicrocontroller

This answer is correct.

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QUESTION 8 OF 102

What electronic control switch will you use to automate the switching on and off of a high current and high power electrical circuits like that of a bioprocessing plant?

Choose only ONE best answer.

A

Circuit breaker

✗

B

Metal-oxide-semiconductor field-effect transistor (MOSFET)

✓

C

Power switch

D

Power transistor

This answer is incorrect. The correct answer is 'B'.

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QUESTION 9 OF 102

Evaluate and recommend the most appropriate sensor to be used in a microcontroller for automating the turning on and off of farmstead street lighting system based on the presence of daylight. What sensor do you recommend?

Choose only ONE best answer.

A

Photosensor

B

Infrared sensor

C

Photovoltaic sensor

D

Ultrasonic sensor

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QUESTION 10 OF 102

States that the voltage or potential difference between two points is directly proportional to the current or electricity passing through the resistance and directly proportional to the resistance of the circuit.

Choose only ONE best answer.

A Ohm's law

B Voltage law

C Amperage Law

D Electrical Law

This answer is correct.

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QUESTION 11 OF 102

Based on the 2009 Philippine Electrical Code, what is the minimum conductor size in mm² for a voltage rating of 0-2,000 volts when using copper wire?

Choose only ONE best answer.

A

2 ✓

B

6

C

8

D

10 ✗

This answer is incorrect. The correct answer is 'A'.

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A farmhouse requires a minimum conductor size of 2 mm² which is equivalent to AWG ____.

Choose only ONE best answer.

A10

B12

C14

D16

This answer is incorrect. The correct answer is 'C'.

AWG 14 has 2.08 mm² conductor size, AWG 16 should not be used in houses.

Except for the drop wire, the common household wires are AWG nos. 10, 12 & 14.

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QUESTION 13 OF 102

Which electrical wire in the list below has the biggest cross-sectional area?

Choose only ONE best answer.

A

AWG 14

B

AWG 12

C

AWG 10

D

AWG 8

This answer is correct.

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QUESTION 14 OF 102

Which type of electrical wire has a maximum operating temperature of 90 °C?

Choose only ONE best answer.

A

THW

B

RHW

C

THHN

D

THWN

This answer is correct.

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QUESTION 15 OF 102

Based on the Phil. Electrical Code, what is the minimum conductor size in mm² for a voltage rating of 2,001-8,000 volts when using copper, aluminum or copper-clad aluminum wire?

Choose only ONE best answer.

A2

B6

C8

D10

This answer is incorrect. The correct answer is 'C'.

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QUESTION 16 OF 102

If V is voltage, I is current and P is power, according to Ohm's law, V is equal to _____.

Choose only ONE best answer.

A

P/I

B

IR

This answer is correct.

C

I/P

D

I^2R

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QUESTION 17 OF 102

Ohm's law is applicable to ____.

Choose only ONE best answer.

A

DC only

B

AC only

C

Both DC and AC

D

DC but not AC

This answer is correct.

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QUESTION 18 OF 102

Ohm's law applies to both AC and DC circuits as long as mutual _____ of the circuit is considered.

Choose only ONE best answer.

A

Resistance

B

Capacitance

C

Conductance

D

Inductance

This answer is incorrect. The correct answer is 'D'.

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The resistivity of annealed copper in ohm meters at 20°C.

A 1.72×10^{-8}

B 1.59×10^{-8}

C 1.68×10^{-8}

D 2.65×10^{-8}

This answer is correct.

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QUESTION 20 OF 102

The resistance of a certain length of wire can be obtained by multiplying its _____ with its length and dividing the resulting product by the cross-sectional area of the wire.

Choose only ONE best answer.

AConductivity

BResistivity

CMalleability

DDuctility

This answer is correct.

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QUESTION 21 OF 102

A farmstead has its farthest load located 350 linear meters from a single phase power source. What length of the wire in meters should be used in the voltage loss computation?

Choose only ONE best answer.

A

350

B

600

C

700 ✓

D

500

This answer is correct.

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In connecting electricity from a post to a farmhouse, a _____ wire is needed.

A Transmission

B Drop

C Automotive

D Electrical

This answer is correct.

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The minimum wire size for a farmhouse drop wire.

A AWG 4

B AWG 6

C AWG 8

D AWG 10

This answer is correct.

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A solar powered slaughterhouse uses electrical stunning gun to make the animals insensible to pain before the slaughter. What should be the voltage of the stunning gun?

Choose only ONE best answer.

- D** 220 Vac

This answer is correct.

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QUESTION 25 OF 102

Based on PAES 418:2002, the recommended lighting intensity in lux (lumen/m²) for fruit and vegetable processing plant are: 100 for outside the building, 200 for general indoor locations, and ___ for working tables.

Choose only ONE best answer.

A

300

B

500

✓

C

400

D

600

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Choose only ONE best answer.

C 75

D 60

This answer is correct.

Since a 100-watt daylight-color LED bulb brightens by about 1,600 lumens

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QUESTION 27 OF 102

What is the voltage needed to light a 1-ampere LED bulb having a resistance of 220 ohms?

Choose only ONE best answer.

A1 volt

B220 volts

C2 volts

D110 volts

This answer is correct.

$P = \text{power}$

$I^2 \times R$

$\frac{V^2}{R}$

$\frac{V}{R}$

$I = \text{current}$

$\frac{P}{V}$

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This answer is correct.

The formula wheel is a circular diagram divided into four quadrants, each representing a different electrical quantity. The center contains the four basic units: Watts (P), Amps (I), Volts (V), and Ohms (R). The quadrants are color-coded and labeled as follows:

- Top-Left (Pink):** P = power. Formulas: $I^2 \times R$ and $\frac{V^2}{R}$.
- Top-Right (Yellow):** I = current. Formulas: $\frac{V}{R}$ and $\frac{P}{V}$.
- Bottom-Right (Blue):** R = resistance. Formulas: $\frac{V}{I}$ and $\frac{V^2}{P}$.
- Bottom-Left (Green):** V = voltage. Formulas: $\frac{P}{I}$ and $\sqrt{P \times R}$.

Diagonal formulas also connect the center to the outer edge: $V \times I$ (top-left to bottom-right), $I \times R$ (bottom-left to top-right), and $\sqrt{P \div R}$ (top-right to bottom-left).

Applying the formula $V=IR$ in the third quadrant of the formula wheel,

$$\begin{aligned} V &= I \times R \\ &= (1 \text{ A}) \times (220 \text{ ohms}) \\ &= 220 \text{ volts} \end{aligned}$$

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How many amperes will pass through a circuit having ten 22-watt LED bulbs if the source is 220 volts?

Choose only ONE best answer.

A

1 Amp

B

220 Amp

C

2 Amp

D

110 Amp

This answer is correct.

Applying the formula $I=P/V$ in the fourth quadrant of the formula wheel,

$$\begin{aligned} I &= P/V \\ &= (10 \times 22 \text{ w}) / (220 \text{ v}) \\ &= 1 \text{ Amp} \end{aligned}$$

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What is the electrical energy consumption of a 15-kw vermi cast production shredder plugged in a 220-volt source if it requires 2 hours to shred 1.5 tonnes of farm waste?

Choose only ONE best answer.

A

0.02 kWh/kg

B

0.04 kWh/kg

C

0.06 kwh/kg

D

0.08 kwh/kg

This answer is correct.

$$\begin{aligned} E_c &= P_c T_o / W_i \\ &= 15 \text{ kW } (2 \text{ h}) / 1,500 \text{ kg} \\ &= 0.02 \text{ kWh/kg} \end{aligned}$$

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QUESTION 30 OF 102

Solid annealed copper has a resistivity of 1.72×10^{-8} ohm-meter at 20°C . What is the resistance in ohm-meter per linear meter of AWG 10 solid annealed copper wire having a diameter of 2.588 mm?

Choose only ONE best answer.

A 3.27×10^{-3} ohms

B 2.6×10^{-3} ohms

C 5.5×10^{-3} ohms

D 3.72×10^{-3} ohms

This answer is correct.

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B

$2.6 \times 10^{-3} \text{ ohms}$

C

$5.5 \times 10^{-3} \text{ ohms}$

D

$3.72 \times 10^{-3} \text{ ohms}$

This answer is correct.

Determining the cross-sectional area of the wire in square meters,

$$A = 3.1416 \times (\text{Diameter})^2 / 4$$
$$= 3.1416 \times (2.588 \text{ mm} \times 1 \text{ m} / 1000 \text{ mm})^2 / 4$$
$$= 5.26 \times 10^{-6} \text{ m}^2$$

Determining the resistance of AWG 10 solid annealed copper wire per linear meter,

$$R = \text{Resistivity} \times \text{Length} / \text{Cross-sectional area}$$
$$= (1.72 \times 10^{-8} \text{ ohm-m}) \times (1 \text{ m}) / (5.26 \times 10^{-6} \text{ m}^2)$$
$$= 3.27 \times 10^{-3} \text{ ohms}$$

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QUESTION 32 OF 102

In off-grid farmhouse electrification, which one has the lowest operation and maintenance cost?

Choose only ONE best answer.

A

Solar Home System

B

Hydro power plant

C

Diesel generator

D

Grid extension

This answer is correct.

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QUESTION 33 OF 102

Which one has the highest operation and maintenance cost?

Choose only ONE best answer.

A

Solar Home System

B

Hydro power plant

C

Diesel generator

D

Grid extension

This answer is correct.

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QUESTION 34 OF 102

In off-grid barangay electrification located 3 kilometers from the electricity grid, which one is recommended?

Choose only ONE best answer.

A

Solar Home System

B

Hydro power plant

C

Diesel generator

D

Grid extension

This answer is correct.

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QUESTION 35 OF 102

Which electricity source has no greenhouse gas emission?

Choose only ONE best answer.

A Solar Home System



B Hydro power plant

C Diesel generator

D Grid extension

This answer is correct.

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Which electricity source has the highest greenhouse gas emission?

A Solar Home System

B Hydro power plant

C Diesel generator

D Grid extension

This answer is correct.

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QUESTION 37 OF 102

To minimize power loss, which equipment is needed to transform DC electricity from a centralized solar power plant into AC electricity before transmitting it to distributed 220-volt farmstead electrical devices?

Choose only ONE best answer.

- A Inverter ✓
- B Transformer
- C Converter
- D Transmitter

This answer is correct.

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QUESTION 38 OF 102

A 950-kW cassava flour processing plant is to be powered using hydro. Which electricity source will you recommend?

Choose only ONE best answer.

A

Picohydro

B

Microhydro

C

Minihydro

D

Small hydro

This answer is correct.

Under the Philippine classification, minihydro is over 100 kW up to 10 MW

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A 3-hectare hillside farm is to be energized. It is 20 meters below a river having a dependable summer discharge of 50 liters/second. Which electricity source will you recommend?

Choose only ONE best answer.

A

Picohydro

B

Microhydro

C

Minihydro

D

Small hydro

This answer is correct.

Picohydro is up to 5 kW only while mini and regular hydros are oversized

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The 1-kW lighting of a farmstead is to be energized using hydropower. It is 10 meters below a creek having a dependable summer discharge of 5 liters/second. Which hydropower will you recommend for development?

Choose only ONE best answer.

A

Picohydro

B

Microhydro

C

Minihydro

D

Small hydro

This answer is correct.

Since power requirement is less than 5 kw, picohydro is recommended.

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QUESTION 41 OF 102

An 8-bladed horizontal axis windmill with 12-meter tall tower has 2.5-m diameter rotor. Estimate the power output in watts using the rule of thumb formula. Average windspeed in the site is 3.2 m/s.

Choose only ONE best answer.

A61

B17

C71

D16

This answer is correct.

Using the rule of thumb formula, this takes efficiency into consideration

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A61

B17

C71

D16

This answer is correct.

Using the rule of thumb formula, this takes efficiency into consideration

$$P = 0.1 AV^3$$
$$P = 0.1 [3.14 \times (2.5 \text{ m})^2 / 4] [3.2 \text{ m/s}]^3$$
$$= 16.08 \text{ watts}$$

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How many households can be powered by a minihydro having a power generation output of 500 kW if an average household consumes 1 kw and power loss is 15%?

Choose only ONE best answer.

A245

B425

C452

D524

This answer is correct.

No. of households served
= (Power Output – Power Loss) / Household consumption
= (500 kw – 0.15 (500 kw) / 1 kW
= 425 households

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0 DISCUSSIONS

Choose only ONE best answer.



C 2

D 15

No. of batteries in series
= System voltage / Battery nominal voltage
= 48 volts / 12 volts
= 4 batteries in series

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 QUIZ - 0 QUESTION

Items, 5 hours Time Limit, 85% Passing

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 QUIZ · 0 QUESTION

Items, 5 hours Time Limit, 85% Passing

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Items, 5 hours Time Limit (5.5 hours for 100

items in the actual board exam), 85%

Passing

 QUIZ • 103 QUESTIONS

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0 DISCUSSIONS

Choose only ONE best answer.

D 15

No. of batteries in series
 = System voltage / Battery nominal voltage
 = 48 volts / 12 volts
 = 4 batteries in series

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0 DISCUSSIONS

D 15

No. of batteries in parallel
= Total number of batteries / No. of batteries in series
= 60 total batteries / 4 batteries in series
= 15 batteries in parallel

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QUIZ · 100 QUESTIONS

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How many kilowatts can be generated from a waterfall having a discharge of $1 \text{ m}^3/\text{s}$, reliable head of 100 meters during summers and system efficiency of 60%?

Choose only ONE best answer.

A

588

B

598

C

589

D

859

This answer is correct.

$$P = 9.81 QHE, \quad P \text{ in kw, } Q \text{ in } \text{m}^3/\text{s}, H \text{ in m, } E \text{ in ratio}$$
$$= 9.81 (1 \text{ m}^3/\text{s}) (100 \text{ m}) (0.6)$$
$$= 588.6 \text{ kW}$$

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QUESTION 46 OF 102

A hydropower resource with water outlet 95 meters high from ground surface has fluctuating discharge of $1.8 \text{ m}^3/\text{s}$ during summers and $3.5 \text{ m}^3/\text{s}$ during rainy months. How many megawatts can be generated at the powerhouse placed below the ground surface with the turbine water inlet 30 meters below ground level? Assume a reliable system efficiency of 60%.

Choose only ONE best answer.

A

1.32

✓

B

1.23

C

2.13

D

3.12

This answer is correct

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0 DISCUSSIONS

- A 1.32 ✓
- B 1.23
- C 2.13
- D 3.12

This answer is correct.

Use the minimum discharge, include depth of turbine water inlet to the head.

$$\begin{aligned} P &= 9.81 QHE, \quad P \text{ in kW, } Q \text{ in m}^3/\text{s, } H \text{ in m, } E \text{ in ratio} \\ &= 9.81 (1.8 \text{ m}^3/\text{s}) (95 \text{ m} + 30 \text{ m}) (0.6) \\ &= 1,324 \text{ kW} \times 1 \text{ MW}/1,000 \text{ kW} \\ &= 1.32 \text{ MW} \end{aligned}$$

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Choose only ONE best answer.

A28

B29

C30

D31

This answer is correct.

$$P = \text{No. of modules} \times \text{Power rating} \times \text{Sunlight hours} \times \text{Inversion eff} \times \text{Match eff} \times \text{Trans. eff}$$
$$= 100 \times 0.075 \text{ kW} \times 5 \text{ hours} \times 0.85 \times 0.95 \times 0.95$$
$$= 28.77 \text{ kWh}$$

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Find the dependable power output of a 12-volt DC motor if it is rated at 10 Amperes.

Assume a system efficiency of 80%.

Choose only ONE best answer.

A

120 Watts

B

96 Watts

C

690 Watts

D

600 Watts

This answer is correct.

Form the theoretical formula $P = VI$, multiply VI by efficiency to get output

$$P \text{ output} = \text{Voltage} \times \text{Amperage} \times \text{System efficiency}$$
$$= 12 \text{ V} \times 10 \text{ A} \times 0.8$$
$$= 96 \text{ Watts}$$

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Find the power that should be supplied to a 24-volt DC motor system having a total amperage of 20 Amperes. System efficiency is estimated at 80%.

Choose only ONE best answer.

A

120 Watts

B

96 Watts

C

690 Watts

D

600 Watts

This answer is correct.

Form the theoretical formula $P = VI$, divide VI by efficiency to get input

Power input = Voltage x Amperage / System efficiency

= 24 V x 20 A / 0.8

= 600 Watts

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QUIZ · 102 QUESTIONS

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QUESTION 50 OF 102

Evaluate and quantify the surge current of a 24-volt DC motor if it is supplying current to a 528-Watt, 24-Vdc water pump. Surge current is estimated at 25%.

Choose only ONE best answer.

A

27.5 A

✓

B

27.2 A

C

2.27 A

D

22 A

This answer is correct.

Since Surge Current is about 25%

$$P = VI$$
$$I = P/V$$
$$= 528/24$$

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D 22 A

$$\begin{aligned}\text{Surge Current} &= (1 + \text{Surge current ratio}) \times \text{Load Amperage} \\ &= (1 + 0.25) \times (22 \text{ Amp}) \\ &= 27.5 \text{ Amp}\end{aligned}$$

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QUESTION 51 OF 102

Applying the principles of electronics, what control switch will you use to automate the switching on and off of a high current and high power electrical circuits like that of a farmstead?

Choose only ONE best answer.

A

Circuit breaker

B

Power switch

C

Power transistor

D

Metal-oxide-semiconductor field-effect transistor (MOSFET)

✓

This answer is correct.

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QUESTION 52 OF 102

If you are to carry out the planning of an automated control for the switching on and off of the electricity supply of an AC powered central pivot sprinkler system, what should be the central controller of your system?

Choose only ONE best answer.

A

Computer

B

Microcontroller

C

Irrigation controller

D

Central pivot controller

This answer is correct.

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You are to evaluate and recommend the most appropriate sensor to be used in a microcontroller for automating the turning on and off of the lighting system of a farmstead based on the presence of sunlight. What sensor will you recommend?

A Ultrasonic sensor

B Infrared sensor

C Photosensor

D Photovoltaic sensor

This answer is correct.

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QUESTION 54 OF 102

The electronic board which is able to read inputs like light on a sensor, temperature reading, a finger on a button, or a social media message and can turn these inputs into outputs like activating a motor, turning on an LED, turning on a solenoid valve, or publishing something online.

Choose only ONE best answer.

A

Microcontroller

B

Integrated circuit

C

Microchip

D

Switch board

This answer is correct.

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A microcontroller is used to activate mist sprinkler system for 25 minutes every time the greenhouse temperature rises above a certain level. What device is best turned on by the microcontroller to start the mist sprinklers?

A Mist sprinkler pump

B Gate valve

C Solenoid valve

D Power switch

This answer is correct.

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QUESTION 56 OF 102

The name given to a set of codes for a commonly used microcontroller program which serve as instructions to the microcontroller.

Choose only ONE best answer.

AProgram

BSketch

CControl

DApp

This answer is correct.

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QUESTION 58 OF 102

A diode needed as protection against lightning in solar panels.

Choose only ONE best answer.

A

Bypass diode

B

Shading diode

C

Blocking diode

D

Lightning diode

This answer is correct.

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QUESTION 59 OF 102

The common pin integrated circuits, made as the pair to other pins when connecting wires in microcontrollers.

Choose only ONE best answer.

A+12 Vdc input

B+5 Vdc output

CGround

DBinary Output

This answer is correct.

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An integrated circuit (IC) that is included as an important part of electronic instrumentation for gathering time and time-based data like velocity, speed, discharge and acceleration.

A Voltage reducer

B Variable resistor

C Timer

D Sensor

This answer is correct.

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QUESTION 61 OF 102

Which one is a single timer integrated circuit that is appropriate in simple instrumentation and control?

Choose only ONE best answer.

- A** IC 555 ✖
- B IC 714 ✔
- C IC 565
- D IC 558

This answer is incorrect. The correct answer is 'B'.

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0 DISCUSSIONS

QUESTION 62 OF 102

Which one is an obsolete timer integrated circuit that should not be used in your new instrumentation and control projects?

Choose only ONE best answer.

A

IC 555

B

IC 741

C

IC 556

D

IC 558

This answer is incorrect. The correct answer is 'D'.

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0 DISCUSSIONS

QUESTION 63 OF 102

What voltage is appropriate in remote instrumentation and control?

Choose only ONE best answer.

- A** 9, 7, and 5 Vdc ✓
- B 7 Vdc
- C 9 Vdc
- D 5 Vdc

This answer is correct.

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0 DISCUSSIONS

In automatic control of irrigation dam gates, what switch is useful in the instrumentation?

A Automatic transfer switch (ATS)

B Pressure switch

C Power switch

D Time switch

This answer is incorrect. The correct answer is 'A' .

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0 DISCUSSIONS

QUESTION 66 OF 102

Switch for automatic pump motor control of sealed water tanks in drip irrigation systems.

Choose only ONE best answer.

A

Automatic transfer switch

B

Pressure switch

C

Power switch

D

Time switch

This answer is correct.

NEXT

0 DISCUSSIONS

In a continuous-fed floating-type biogas digester, the upper end of the outlet pipe should be _____ with in the upper end on the inlet pipe.

A Lower

B higher

C Level

D lower by 10%

This answer is incorrect. The correct answer is 'C'.

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QUESTION 67 OF 102

In a continuous-fed floating-type biogas digester, the upper end of the outlet pipe should be _____ with in the upper end on the inlet pipe.

Choose only ONE best answer.

A Lower

B higher

C Level

D lower by 10%



This answer is incorrect. The correct answer is 'C' .

See Figure 3. PAES 413:2001



0 DISCUSSIONS

Determine which of the following floating-type rectangular biogas digesters with integrated digester and gasholder follows the standard height/length ratio for optimum design based on PAES 413:2001?

A Length is 1.2 times the side, height side ratio of 0.5

B Length is thrice the side, height side ratio of 0.32

C Length is twice the side, height side ratio of 0.34

D Length is 4 times the side, height side ratio of 0.31

This answer is incorrect. The correct answer is 'C'.

See Table 3 of PAES 413:2001

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QUESTION 69 OF 102

You are tasked to evaluate the effluent of a rice mill. In addition to high concentration of organic and inorganic substances leading to significant source of pollution, its effluent is also high in _____.

Choose only ONE best answer.

- A** BOD, COD and TDS ✓
- B Chemical oxygen demand (COD)
- C Total Dissolved solid (TDS)
- D Biological oxygen demand (BOD)

This answer is correct.

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A medium-size floating-type biogas digester accommodates manure from 551 heads of porkers of mixed ages and retains manure inside the digester for 30 days. The mean daily manure production of porkers is 2.2 kg/head while the specific gas production for 30-day manure retention period is 0.063 m³/kg. Water:manure ratio is 1:1. What is the daily biogas production?

Choose only ONE best answer.

A 153 m³/day

B 76 m³/day

C 125 m³/day

D 193 m³/day

This answer is correct.

$$P = NMG$$



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Feedlot cattle are used as source of manure to generate 50.1 m^3 of biogas daily.

Retention period is 25 days and the water-manure ratio is 1:1. The mean daily manure production of feedlot cattle is 13 kg/head while its specific gas production for 25-day manure retention period is half of that of chicken dung. Find the number of animal heads required.

Choose only ONE best answer.

A

258 heads

B

129 heads

C

206 heads

D

192 heads

This answer is correct.

$$N = P/(MG)$$

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D 192 heads

$$N = P/(MG)$$

$N = \text{No. of heads}$

C = biogas production

M = manure production,

G = Half of specific gas production of chicken dung

$$= 1/2 \times 0.06 \text{ m}^3/\text{kg} \text{ or } 0.03 \text{ m}^3/\text{kg}$$
$$N = (50.1 \text{ m}^3)/[(13 \text{ kg/head})(0.03 \text{ m}^3/\text{kg})]$$

= 128.46 heads, to avoid gas shortage round up to 129 heads

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A bioprocessing boiler uses 10 units of 50-cm gas burners. The consumption of each burner is 1.19 m³/hr and the boiler is operating 5 hours daily. Find the daily biogas consumption of the whole boiler.

Choose only ONE best answer.

A 82 m^3

B 60 m^3

C 110 m^3

D 95 m^3

This answer is correct.

$$C = NBT$$

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D 95 m^3

$$C = NBT$$

$C = \text{Consumption}$

$N = \text{No. of units}$

B = biogas consumption of device

$T = \text{Time}$

$$C = \text{No. of device units} \times \text{Hourly consumption} \times \text{Time}$$
$$= (10 \text{ units})(1.19 \text{ m}^3/\text{hr})(5 \text{ hrs})$$
$$= 59.5 \text{ m}^3/\text{day}$$

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Choose only ONE best answer.

D Floating type

Reference: Biogas Plant Design by Arthur Tambong, 1992

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QUESTION 74 OF 102

The quick method of farm composting that uses live earthworms and plant biomass.

Choose only ONE best answer.

A

Earthworm and biomass composting

B

Biodegrading

C

Biofertilizer production

D

Vermicomposting

This answer is correct.

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QUESTION 75 OF 102

A kind of liquid fertilizer that is derived by mixing vermi cast and water.

Choose only ONE best answer.

A

Vermi fertilizer

B

Vermi tea

C

Biofertilizer

D

Vermi cast

This answer is correct.

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Giant bamboo is a fast-growing environment friendly tall grass that neutralizes greenhouse gas emissions in the atmosphere. Give the scientific name of a giant bamboo variety that is suited in the Philippines.

A *Dendrocalamos giganteus*

B *Bambosa giganteus*

C *Dendrocalamus asper*

D *Asper giganteus*

This answer is incorrect. The correct answer is 'C'.

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QUESTION 77 OF 102

A new construction material made by processing the raw bamboo culm into a laminated composite material similar to glue-laminated wood products.

Choose only ONE best answer.

A

Plyboo

B

Engineered bamboo ✓

C

Laminated bamboo

D

Processed bamboo

This answer is correct.

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QUESTION 78 OF 102

Out of giant bamboo, you are tasked to manufacture useful construction materials. What product will you manufacture to command high value added?

Choose only ONE best answer.

A Tempered bamboo

B Laminated bamboo

C Engineered bamboo

D Processed bamboo

This answer is correct.

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A construction material similar to plywood made by processing the raw bamboo culm into glued and laminated layers.

Choose only ONE best answer.

A

Engineered bamboo

B

Laminated bamboo

C

Plyboo

D

Processed bamboo

This answer is incorrect. The correct answer is 'C'.

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0 DISCUSSIONS

QUESTION 80 OF 102

A kind of salt that is useful for preserving bamboos.

Choose only ONE best answer.

A

Sea salt

B

Salt lick

C

Boron salt

D

Basalt

This answer is incorrect. The correct answer is 'C' .

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0 DISCUSSIONS

QUESTION 82 OF 102

The toxic component of biogas.

Choose only ONE best answer.

A

Hydrogen sulfide

B

Hydrogen

C

Sludge

D

Sulfur

This answer is correct.

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QUESTION 83 OF 102

The by-product of biogas production commonly used as soil conditioner after air drying.

Choose only ONE best answer.

A Fertilizer

B Waste

C Sludge ✓

D Slurry

This answer is correct.

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QUESTION 84 OF 102

The amount paid to project owners from CDM fund for carbon emission reduction projects.

Choose only ONE best answer.

A

Emission value

B

Carbon credit value

C

Carbon trading value

D

Carbon value

This answer is correct.

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QUESTION 85 OF 102

The acronym of the Inter-Governmental Panel for Climate Change.

Choose only ONE best answer.

A

IGPCC

B

IPC

C

IPCC

✓

D

Inter-GPCC

This answer is correct.

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QUESTION 86 OF 102

Best kind of biogas plant for large animal farms having effluents with low solid content.

Choose only ONE best answer.

A

Fixed dome

B

Covered lagoon

✓

C

Floating type

D

Balloon

This answer is correct.

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QUESTION 87 OF 102

Plastic sheet for constructing balloon and bed lining in covered lagoon biogas plants.

Choose only ONE best answer.

A

High density polyethylene (HDPE)

✓

B

Polyvinyl Chloride (PVC)

C

Acetate

D

UV-stabilized acetate

This answer is correct.

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0 DISCUSSIONS

QUESTION 88 OF 102

By how many times is methane destructive to the environment than carbon dioxide?

Choose only ONE best answer.

A

2 times

B

21 times

✓

C

10 times

D

15 times

This answer is correct.

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0 DISCUSSIONS

QUESTION 89 OF 102

Toxic by-product of geothermal power plants.

Choose only ONE best answer.

A

Hydrogen sulfide

✓

B

Carbon monoxide

C

Carbon dioxide

D

Sulfuric acid

This answer is correct.

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QUESTION 90 OF 102

When using covered lagoon biodigester, what is approximate methane content in biogas?

Choose only ONE best answer.

A

40%

B

70%

✓

C

50%

D

80%

This answer is correct.

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QUESTION 91 OF 102

What is the minimum methane content of biogas in order to operate biogas-fueled generators?

Choose only ONE best answer.

A

50%

B

85%

C

65%

✓

D

90%

This answer is correct.

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QUESTION 93 OF 102

A device to convert biogas to electricity.

Choose only ONE best answer.

AConverter

BMotor

C Electricity generator

DBiogas generator

This answer is incorrect. The correct answer is 'D'.

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QUESTION 94 OF 102

A board constructed across the bottom of the biogas digester to prevent early exit of slurry.

Choose only ONE best answer.

A

Baffle board

B

Wall

C

Divider

D

Slurry retainer

This answer is correct.

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You are tasked to implement a biogas project for a poultry having 5,000 heads of layers.

The prevailing daytime temperature in the project site is 30°C. Which of the biogas digesters below will you use?

Choose only ONE best answer.

A

Fixed type

B

Floating type

C

Tube bag digester

D

Covered lagoon digester

This answer is correct.

Since ambient temperature is in the likelihood of 30°C, it favors floating type digester

Reference: Biogas Plant Design by Arthur Tambong, 1992

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Covered lagoon biodigester, minihydro and solar energy projects are evaluated for financial feasibility. Only one feasible project can be funded. Each of them requires PHP 20,000,000 investment. The financial internal rates of return are 8%, 9% and 10%, respectively. Annual discount plus inflation rate is 11%. Which project shall be implemented?

Choose only ONE best answer.

A

Covered lagoon biodigester

B

Minihydro

C

None

D

Solar

This answer is correct.

Since all their IRRs are less than the discount plus inflation rate

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QUESTION 97 OF 102

In the Clean Development Mechanism (CDM), how much will the project owner of a new covered lagoon biodigester receive annually from CDM fund for the first 10 years if his biodigester produces 8,000 tonnes of methane and 3,000 tonnes of carbon dioxide per year. Assume that the current carbon credit costs US\$8 per tonne of carbon equivalent.

Choose only ONE best answer.

A

US\$ 30,000

B

US\$ 100,000

C

US\$ 1,368,000

D

US\$ 2,200,000

This answer is correct.

Considering that methane is 21 times more destructive than carbon,

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A	US\$ 30,000
---	-------------

B US\$ 100,000

C US\$ 1,368,000

D	US\$ 2,200,000
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Considering that methane is 21 times more destructive than carbon,

$$= \text{Carbon cost} \times [(\text{tonnes carbon/year} + 21 (\text{tonnes of methane/year}))]$$
$$= \text{US\$ } 8/\text{tonne} [3,000 \text{ tonnes} + 21(8,000 \text{ tonnes})]$$
$$= \text{US\$ } 1,368,000/\text{year}$$

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What is the daily biogas consumption in cubic meters of an industrial boiler using 12 units of 50-cm gas burner 8 hours a day? The consumption of each gas burner is 1.3 cu.m./hour.

Choose only ONE best answer.

A

9.6

B

102

C

125

D

80

This answer is correct.

$$C = NBT$$

where C = Consumption, N = No. of units, B = biogas consumption of device, T = Time

$$C = (12 \text{ units})(1.3 \text{ m}^3/\text{hr})(8 \text{ hrs})$$
$$= 124.8 \text{ m}^3$$

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Estimate the biogas production in cubic meters per day from a large balloon type digester using 1:1 water-manure ratio. The source farm has 1,100 heads of porkers of mixed ages. Use a retention period of 30 days. The mean daily manure production of porkers is 2.2 kg/head while the specific gas production for 30-day manure retention period is 0.063 m³/kg.

Choose only ONE best answer.

A

97

B

153

C

125

D

193

This answer is correct.

$$P = NMG$$

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Subject C Preboard Exam Part 1 - Practice

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QUIZ · 100 QUESTIONS

Subject C Preboard Exam Part 2 - Practice

QUIZ · 0 QUESTION

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Subject C Preboard Exam Part 3 - Practice

QUIZ · 0 QUESTION

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Subject C Preboard Exam Part 2 - Practice

0 DISCUSSIONS

A97

B153

C125

D193

This answer is correct.

$$P = NMG$$

where P = biogas production potential, N = Number of heads, M = manure production, G = Specific gas production of the manure for a specific retention period

$$P = (1,100 \text{ heads})(2.2 \text{ kg/head})(0.063 \text{ m}^3/\text{kg})$$
$$= 152.46 \text{ m}^3$$

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QUIZ · 103 QUESTIONS

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Subject C Preboard Exam Part 2 - Practice

QUESTION 100 OF 102

How many heads of breeding cattle are needed as source of manure to generate 80 m³ of biogas a day if the retention period is 25 days and water-manure ratio is 1:1. The mean daily manure production of breeding cattle is 13 kg/head while its specific gas production for 25-day manure retention period is half of that of chicken dung.

Choose only ONE best answer.

A

192

B

205

C

206

D

139

This answer is correct.

$$N = P/(MG)$$

where N = No. of heads, C = biogas production, M = manure production

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Subject C Preboard Exam Part 1 - Practice

Subject C Preboard Exam Part 1 Final - 100
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QUIZ - 100 QUESTIONS

Subject C Preboard Exam Part 2 - Practice

Subject C Preboard Exam Part 2 Final - 102
Items, 5 hours Time Limit, 85% Passing
QUIZ - 102 QUESTIONS

Subject C Preboard Exam Part 3 - Practice

○ Subject C Preboard Exam Part 3 Final - 103 Items, 5 hours Time Limit (5.5 hours for 100 items in the actual board exam), 85% Passing

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Subject C Preboard Exam Part 2 - Practice

0 DISCUSSIONS

Choose only ONE best answer.

A 192

B 205

C 206

D 139

This answer is correct.

$$N = P/(MG)$$

where N = No. of heads, C = biogas production, M = manure production

$G = \text{Half of specific gas production of chicken dung}$
 $= 1/2 \times 0.06 \text{ m}^3/\text{kg} \text{ or } 0.03 \text{ m}^3/\text{kg}$

$$N = (80 \text{ m}^3)/[(13 \text{ kg/head})(0.03 \text{ m}^3/\text{kg})]$$

$$= 205.13 \text{ heads, to avoid gas shortage round up to 206}$$

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Subject C Preboard Exam Part 1 - Practice

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0 DISCUSSIONS

QUESTION 101 OF 102

The biggest biogas digester type in terms of volume that is appropriate for large animal farms.

Choose only ONE best answer.

A

Floating type digester

B

Fixed type digester

C

Tube bag type digester

D

Covered lagoon biodigester

This answer is correct.

NEXT

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0 DISCUSSIONS ↗

QUESTION 102 OF 102

The recommended biogas digester type for most Philippine households.

Choose only ONE best answer.

- A Floating type digester ✓
- B Fixed type digester ✗
- C Tube bag type digester
- D Covered lagoon biodigester

This answer is incorrect. The correct answer is 'A' .

NEXT